



University of Alkafeel, College of Dentistry, Year 3, Course 1

Pathology Lab 10 Infectious Diseases

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Learning Objectives

- ☐ **Define Infectious diseases.**
- ☐ **Classify Infectious agents.**
- ☐ **Know the chain of infection.**
- ☐ **Identify the modes of transmission and the techniques for identifying agents .**
- ☐ **Understand the patterns of tissue injury in infectious agents.**

Introduction

Infectious diseases are disorders caused by pathogenic microorganisms including **bacteria, viruses, fungi, parasites, and prions.**

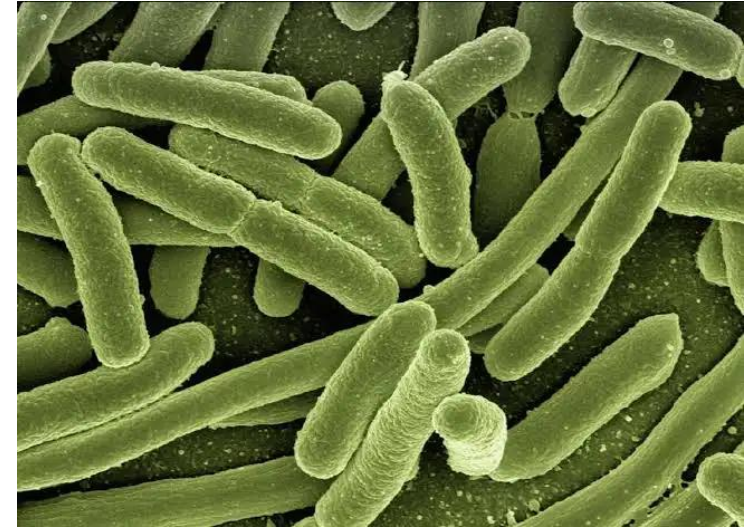
These agents invade host tissues, multiply, and trigger host immune responses that produce clinical disease.

Classification of Infectious Agents

A. Bacteria

Prokaryotic organisms with cell walls.

- Gram-positive (e.g., **Staphylococcus aureus**)
- Gram-negative (e.g., **Escherichia coli**)
- Acid-fast (e.g., **Mycobacterium tuberculosis**)



B. Viruses

Obligate intracellular organisms containing DNA or RNA.

- Example: **Human immunodeficiency virus**
- Example: **Hepatitis B virus**



Classification of Infectious Agents

C. Fungi

Eukaryotic organisms (yeasts, molds).

- Example: **Candida albicans**
- Example: **Aspergillus fumigatus**

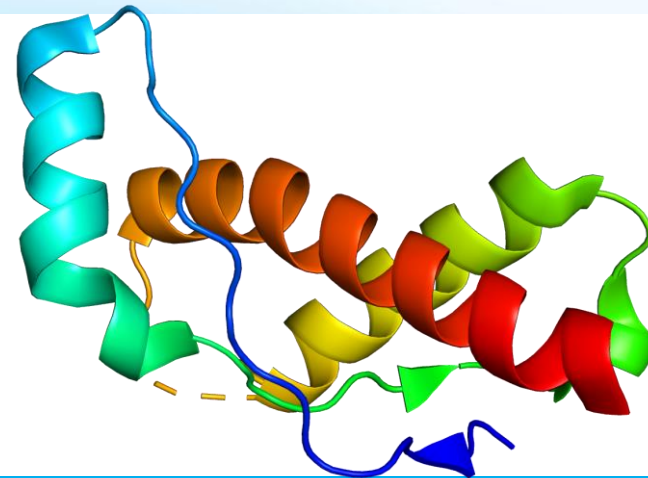
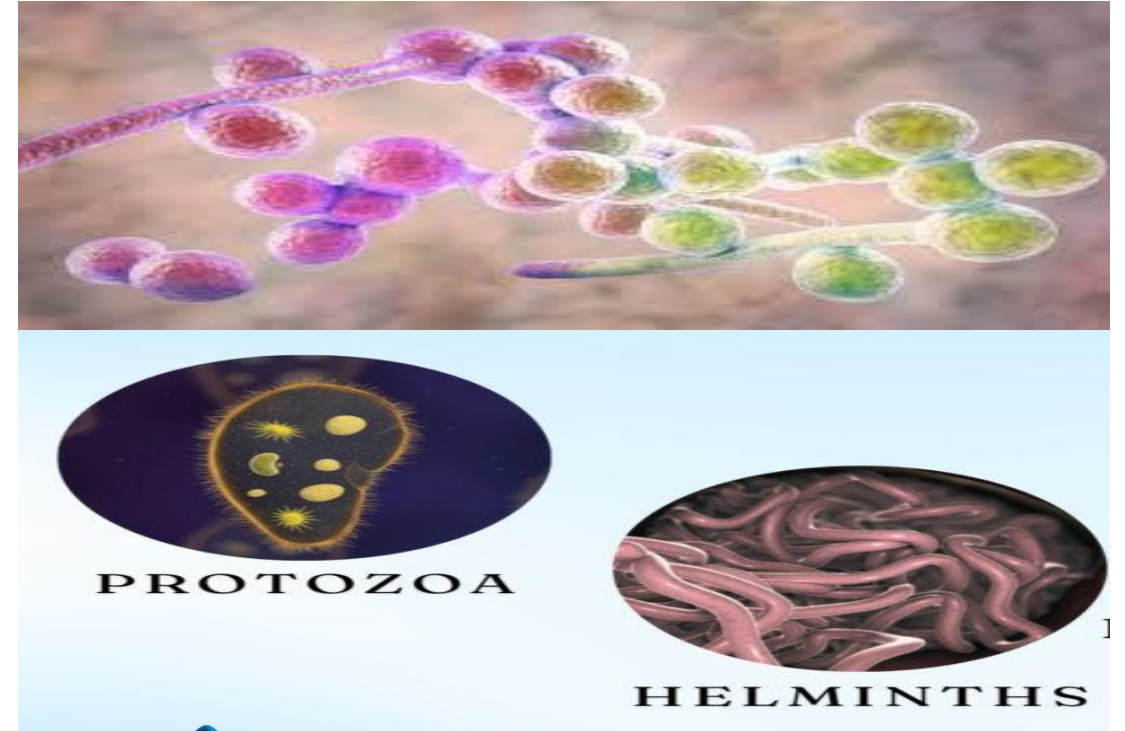
D. Parasites

- Protozoa (e.g., **Plasmodium falciparum**)
- Helminths (e.g., **Ascaris lumbricoides**)

E. Prions

Misfolded proteins causing neurodegenerative disease.

- Example: **Creutzfeldt-Jakob disease**

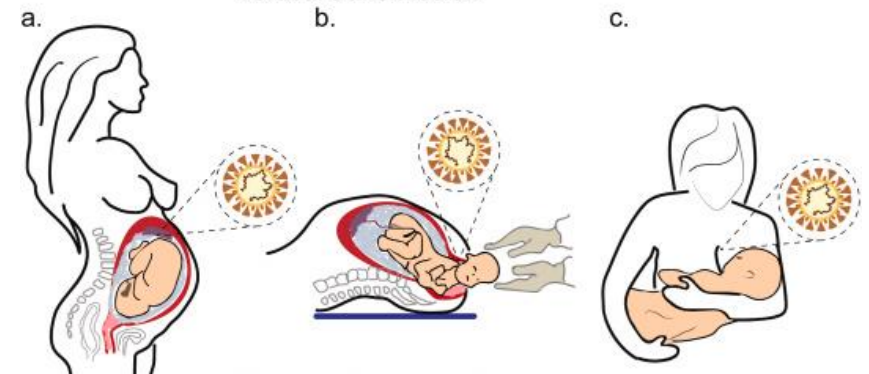
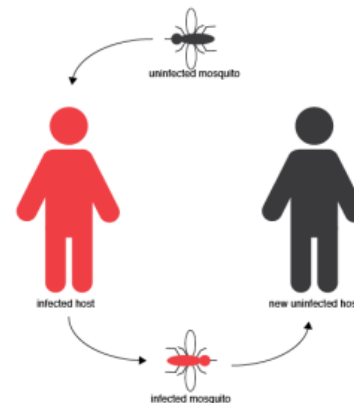
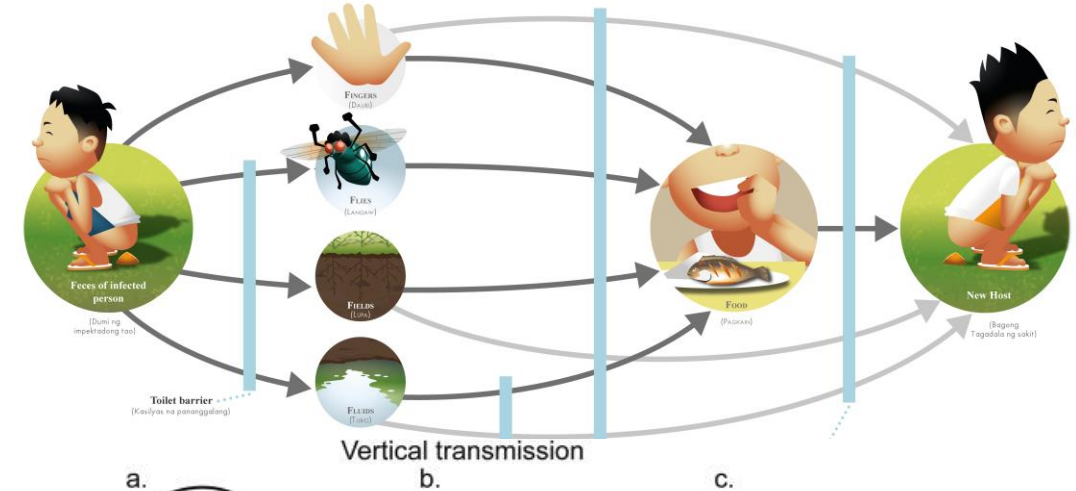
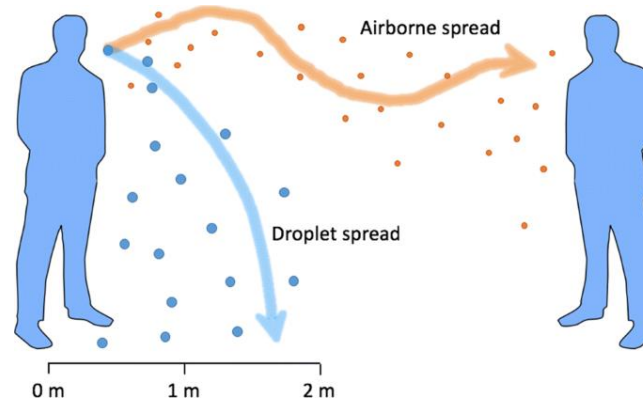


Chain of Infection



Modes of Transmission

- Direct contact (skin-to-skin, sexual)
- Droplet spread
- Airborne
- Fecal-oral
- Vector-borne (e.g., malaria)
- Vertical (mother to fetus)



Techniques for identifying infectious agents

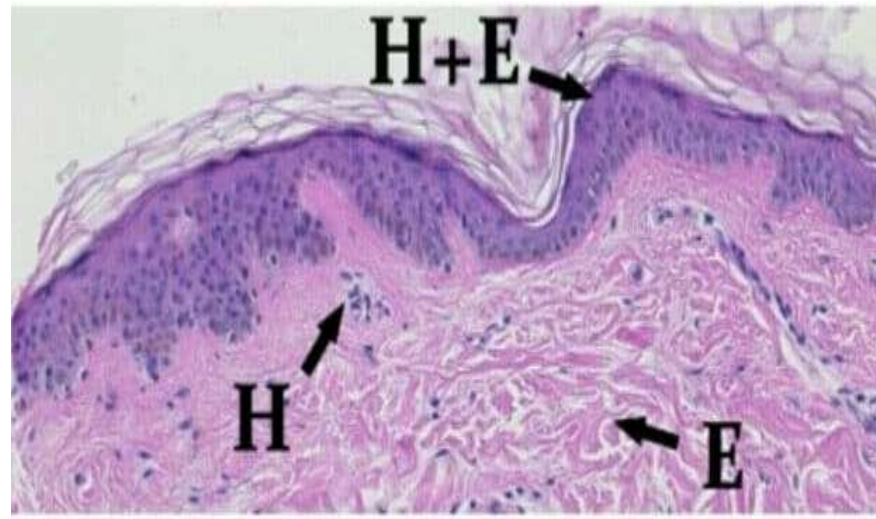
There are several methods for identifying microorganisms in tissue and body fluids :

Culture: bacterial and fungal cultures remain essential for diagnostic testing .



Techniques for identifying infectious agents

Histology : some infectious agents can be seen in hematoxylin and eosin (H&E)–stained sections .

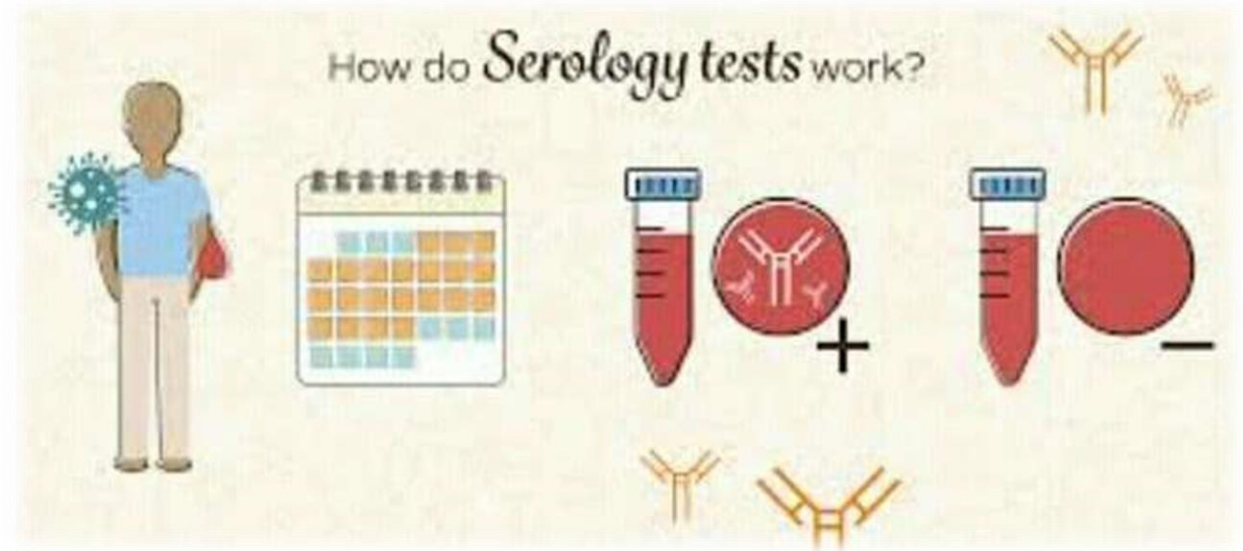


Hematoxylin & Eosin staining



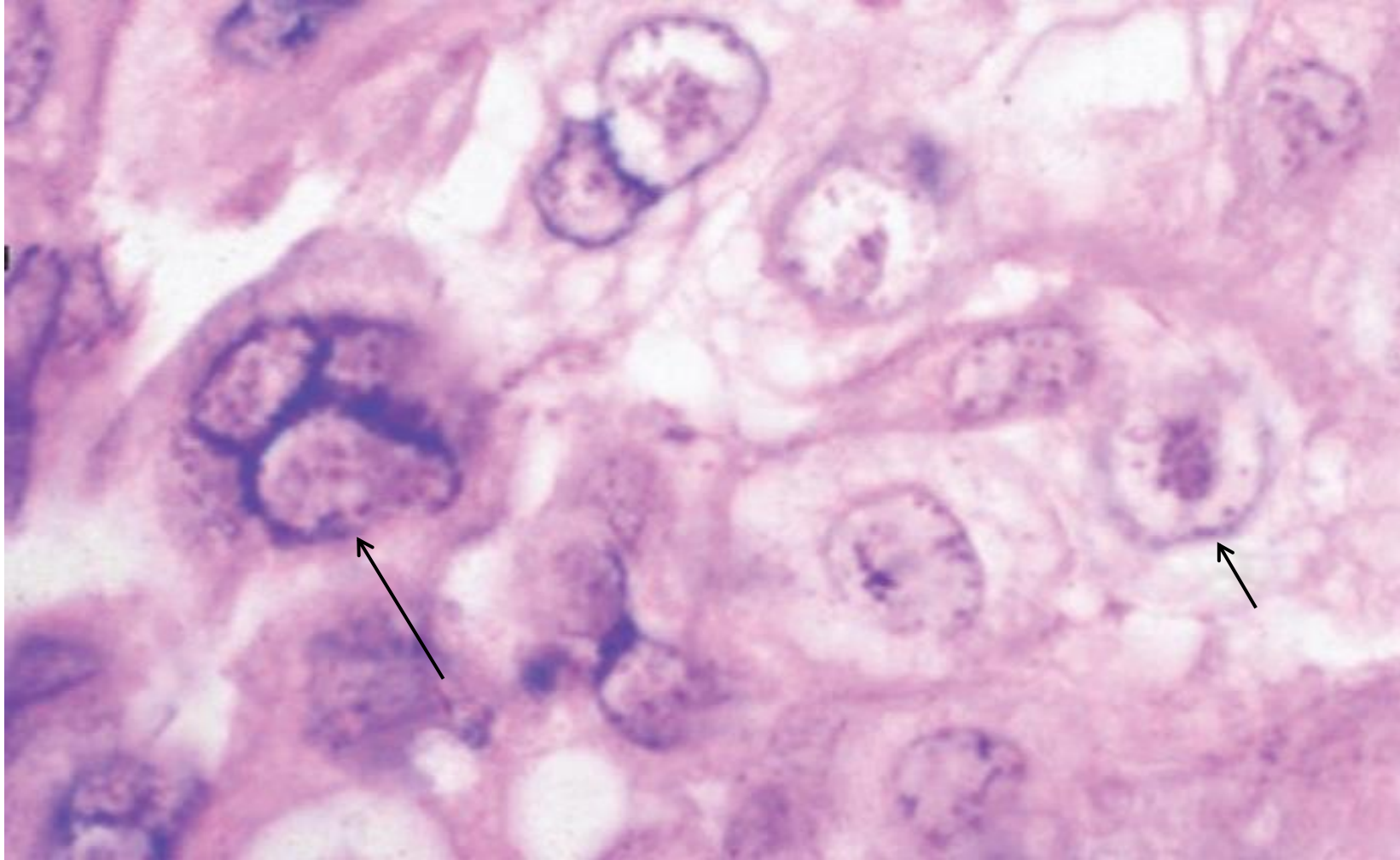
Techniques for identifying infectious agents

Serology : acute infections can be diagnosed serologically by detecting pathogen-specific antibodies in the serum.

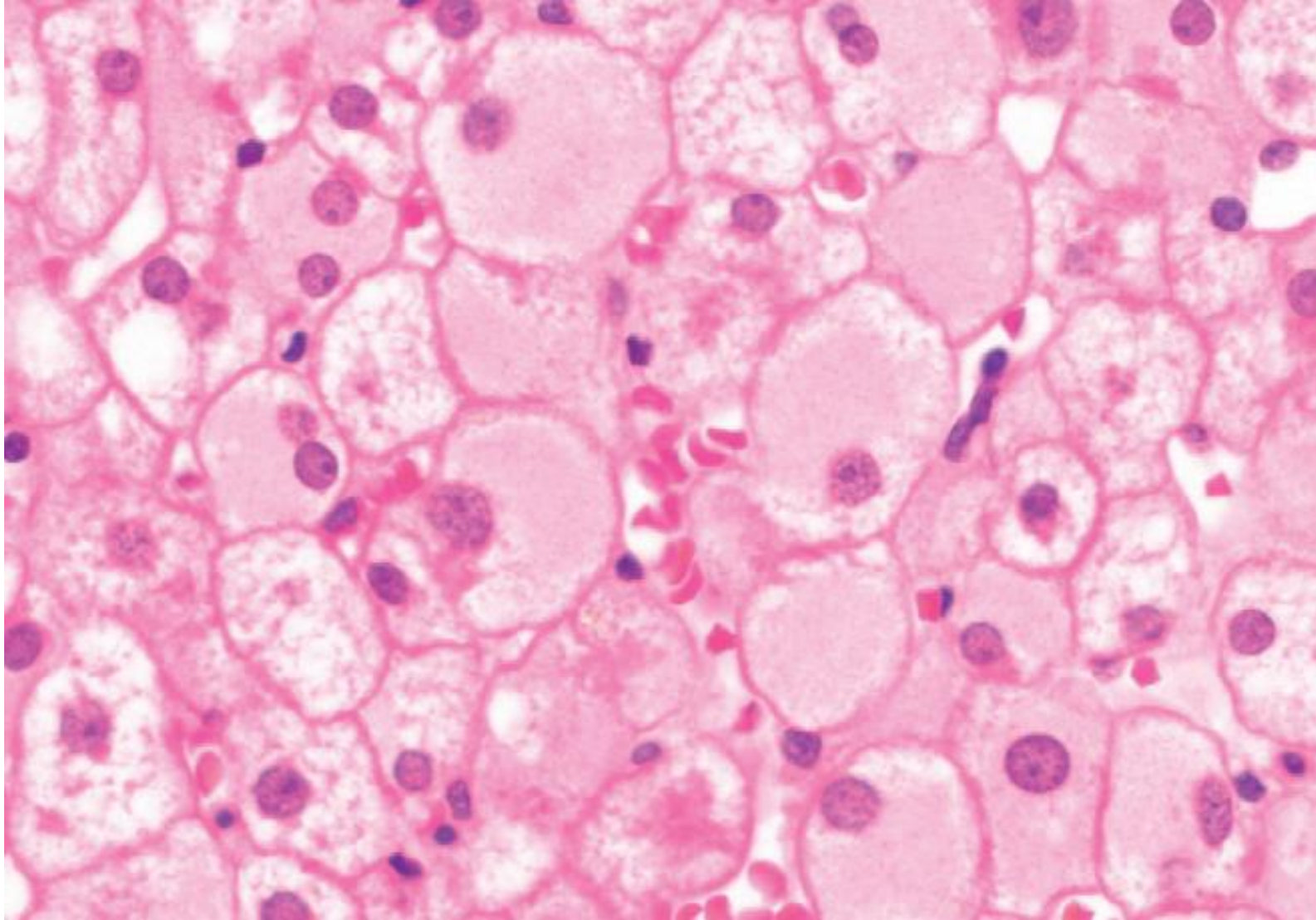




- **Cytomegalovirus infection in the lung. Infected cells show distinct nuclear (long arrow) and ill-defined cytoplasmic(short arrows) inclusions.**



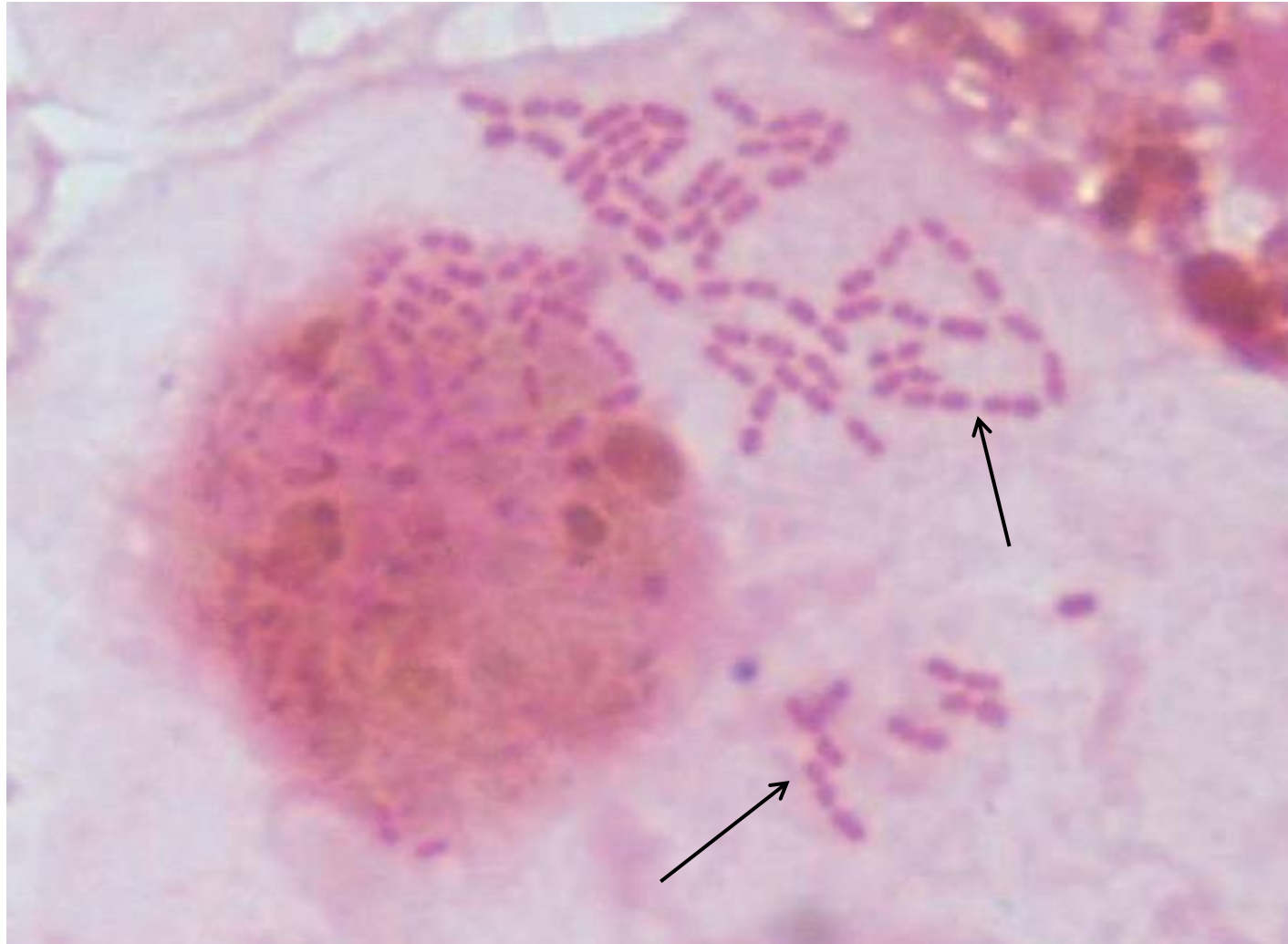
Varicella-zoster virus infection in the skin. Herpes simplex virus and varicella-zoster virus both cause characteristic cytopathologic changes, including fusion of epithelial cells, which produces multinucleate cells with molding of nuclei to one another (long arrow), and eosinophilic haloed nuclear inclusions (short arrow).



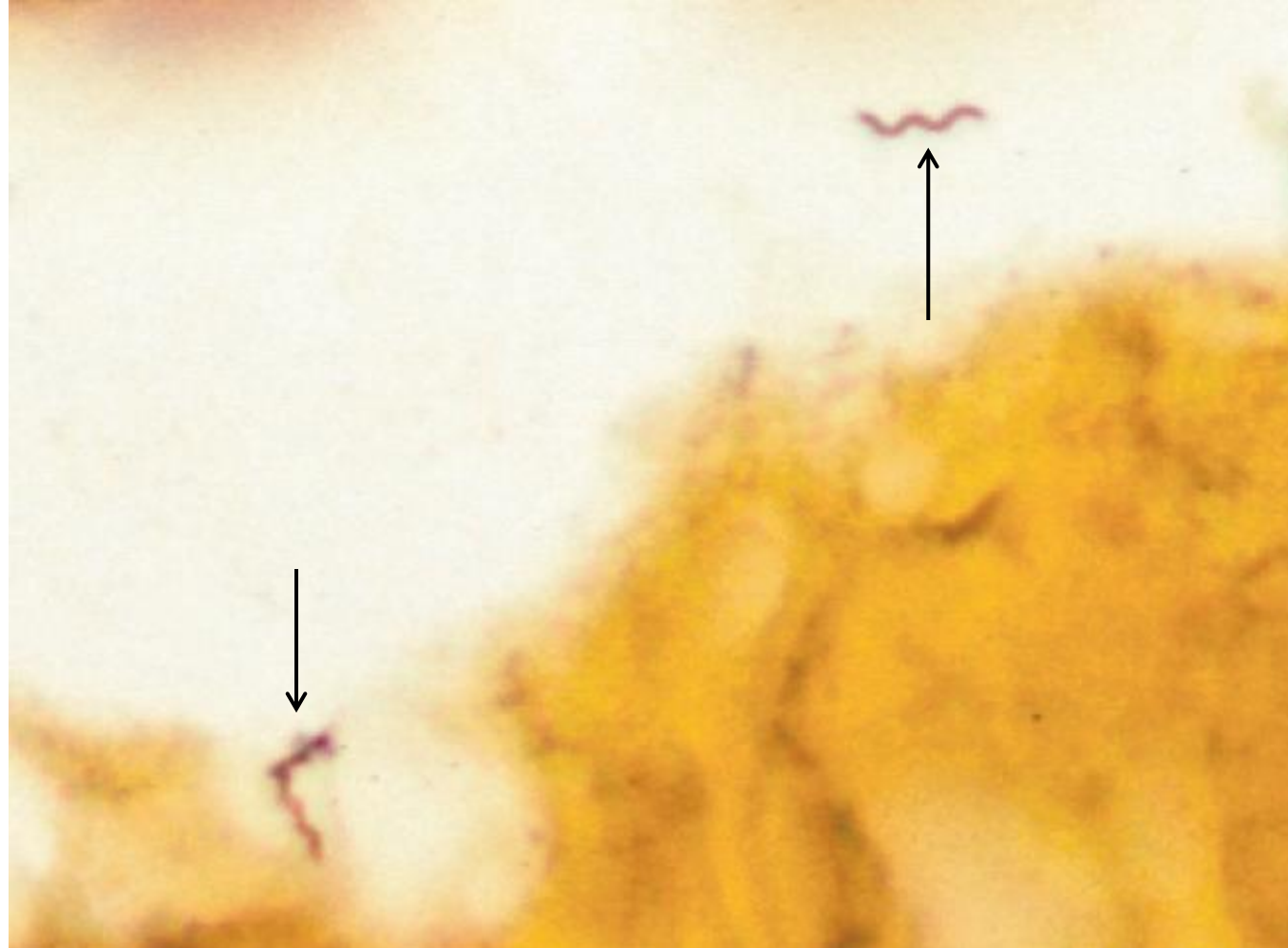
Hepatitis B viral infection in liver. In chronic infections, infected hepatocytes show diffuse granular (“ground-glass”) cytoplasm, reflecting accumulated hepatitis B surface antigen (HBsAg).



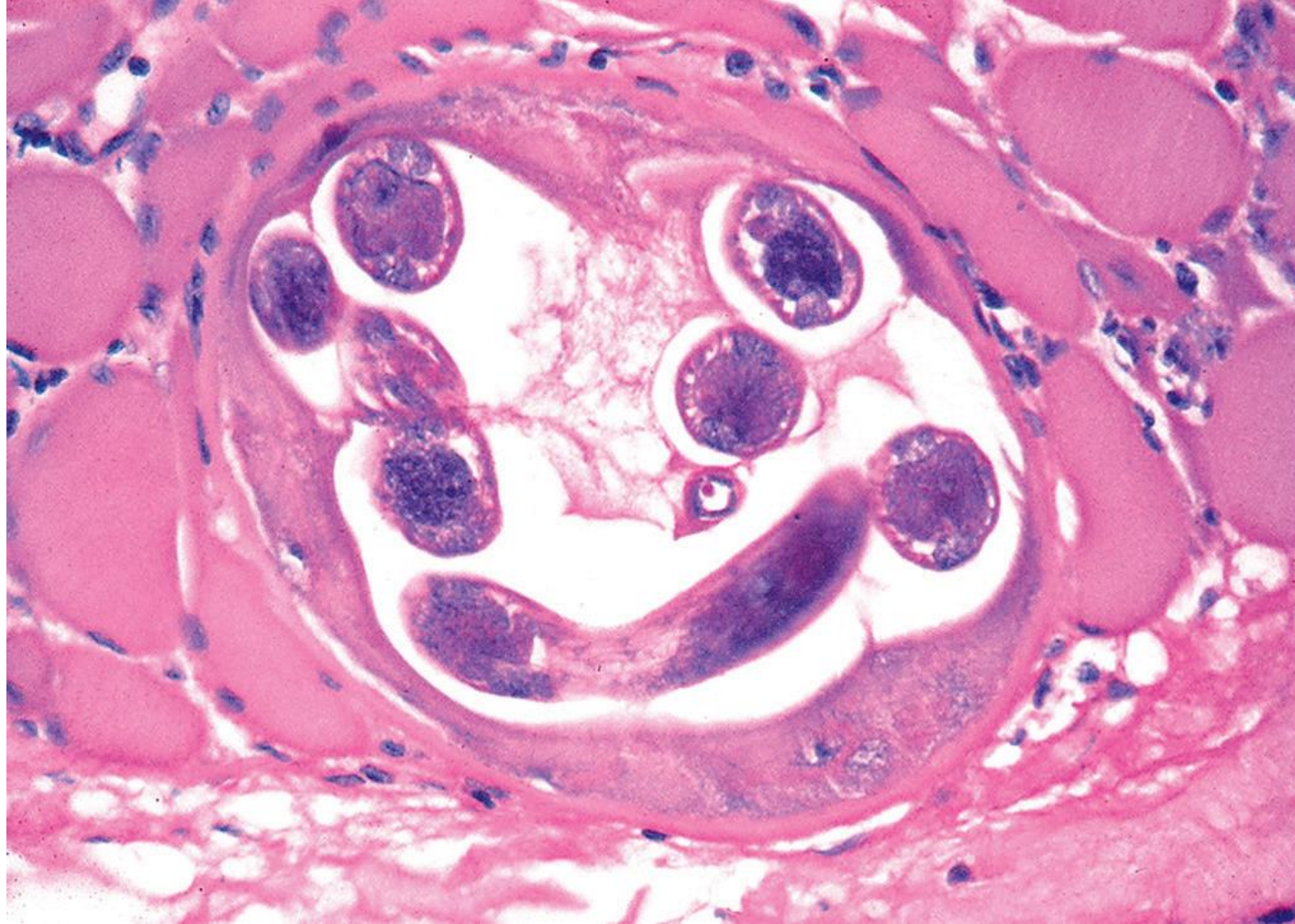
Gram stain preparation of sputum from a patient with pneumonia. Gram-positive, elongated cocci in pairs and short chains (*Streptococcus pneumoniae*) and a neutrophil are evident .



Gram stain preparation of a bronchoalveolar lavage specimen showing gram-negative intracellular rods typical of members of Enterobacteriaceae such as *Klebsiella pneumoniae* or *Escherichia coli*.



Silver stain preparation of brain tissue from a patient with Lyme disease meningoencephalitis .Two helical spirochetes (*Borrelia burgdorferi*) are shown (arrows).



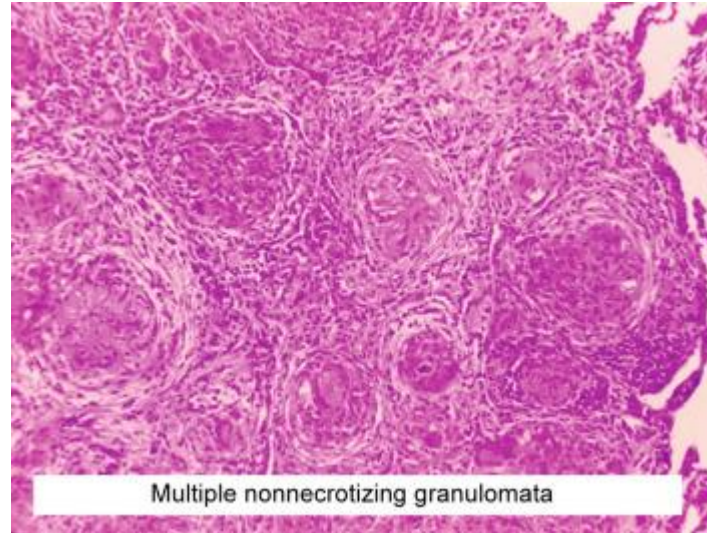
Coiled *Trichinella spiralis* larva within a skeletal muscle cell.

Patterns of Tissue Injury

1. Suppurative (Purulent) Inflammation

- Neutrophil accumulation
- Abscess formation

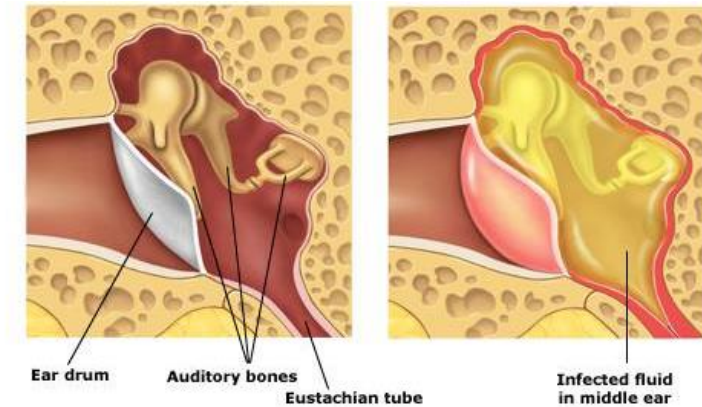
Example: **Staphylococcus aureus**



2. Granulomatous Inflammation

- Epithelioid macrophages
- Giant cells
- Caseous necrosis

Classic example: **Tuberculosis**



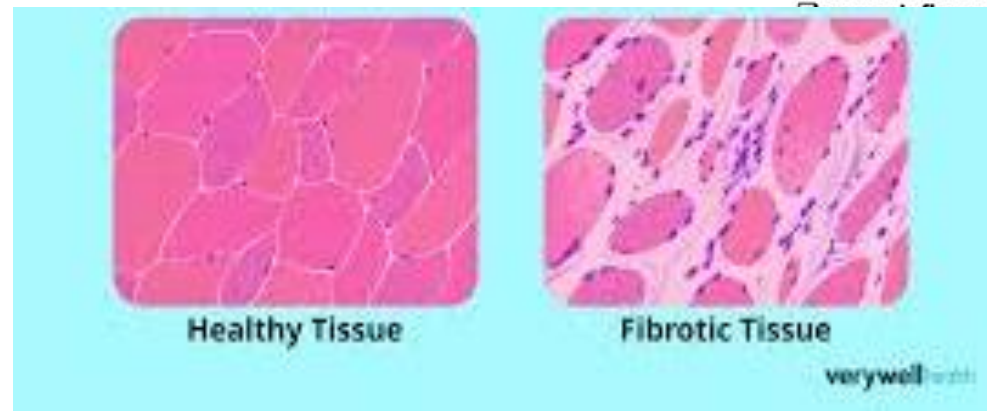
3. Cytopathic–Cytoproliferative Reaction (Viral)

- Inclusion bodies
- Cell necrosis

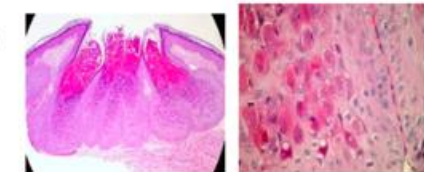
Example: **Herpes simplex virus**

3. Cytopathic–Cytoproliferative Inflammation

- Usually by viruses e.g. Pox virus
- cell necrosis
- cellular proliferation



tory cells
e in the skin
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l, forming



Molluscum Contagiosum (pox virus)

